llmXive Automated Review of Does VLA Even Know the Basics? Measuring Commonsense and World Knowledge Retention in Vision-Language-Action Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong claims regarding the performance gaps between VLMs and VLAs and the specific capabilities of models on the Act2Answer benchmark. While the overall narrative is supported by the data, there are instances where the textual claims are slightly stronger or more absolute than the evidence in the tables permits.

Specifically, in Section 5.3 (RQ3), the authors state that VLMs exceed VLAs by "roughly 20-40 points." While this holds for many categories (e.g., Emotion: 95% vs 48%), it does not hold for others like Color, where the gap is often smaller (e.g., 100% vs 89% for InternVL vs OpenVLA is 11 points). The claim should be qualified to reflect the variance across categories.

More critically, in Section 5.3 (RQ2), the text asserts that "no evaluated VLA reaches above-random performance on Symmetry or Counting." This is factually incorrect based on Table 1. SpatialVLA achieves 52% on Counting, and Xiaomi-Robotics-R0 achieves 58% on Symmetry. While these scores are close to the 50% chance baseline (and likely within the statistical noise defined by Δ in the appendix), the absolute phrasing "no evaluated VLA" is contradicted by the reported numbers. The text should be revised to state that "no VLA demonstrates *robust* or *statistically significant* above-random performance" or to acknowledge the marginal exceptions.

Finally, the claim in RQ5 regarding the superiority of joint-training models is generally supported but presented as a uniform trend. The data shows that while Magma is a clear outlier, other joint-trained models like InternVLA-M1 perform comparably to or sometimes worse than robotics-only baselines in specific categories (e.g., Attribute). The text should avoid implying a strict dichotomy and instead emphasize the "trend" or "average" nature of the finding, as the data shows significant overlap.

These are primarily issues of precision in reporting rather than fundamental flaws in the scientific conclusions. Correcting the absolute phrasing to match the nuance of the data will improve the accuracy of the claims.

Action items.

- **[writing]** In Section 5.3 (RQ3), the claim of a uniform '20-40 points' gap is an overgeneralization. Table 1 shows gaps varying from ~11 points (Color) to ~47 points (Emotion). Please qualify this claim to reflect the variance across categories.
- **[writing]** In Section 5.3 (RQ2), the text states 'no evaluated VLA reaches above-random performance on Symmetry or Counting.' Table 1 shows SpatialVLA at 52% (Counting) and Xiaomi at 58% (Symmetry). While marginal, the absolute claim is factually contradicted. Please revise to 'no VLA demonstrates robust above-random performance' or acknowledge these exceptions.
- **[writing]** In Section 5.3 (RQ5), the text implies a strict performance separation between joint-trained and robotics-only models. Table 1 shows InternVLA-M1 (joint) scoring lower

than OpenVLA (robotics) on Attribute (49% vs 51%). Please qualify the claim to reflect that this is a general trend with exceptions, not a definitive rule.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure presents a radar chart and bar plots comparing VLA models, but the caption's claim of seven models contradicts the five listed in the legend. Additionally, the radar chart does not specify which task suite it represents, and the Act2Answer bars in the bottom panel lack numerical labels.

Figure 2

Figure 2 effectively illustrates the Act2Answer task setup with clear visual examples of the robot arm, cube, and answer plates across diverse categories. The caption accurately describes the task objective, and the embedded text labels within the image panels provide sufficient context for the specific instructions and categories shown.

Figure 3

Figure 3 effectively visualizes the data curation pipeline described in the caption, clearly mapping the flow from diverse VLM benchmarks through LLM processing and human review to the final Act2Answer environment. The diagram is uncluttered, and the visual elements (icons, arrows, and text) are legible and logically organized.

Figure 4

The figure attempts to show probing results for multiple models across tasks, but the legend fails to distinguish between the different models of the same color, rendering the specific performance of individual models unidentifiable.

Figure 5

Figure 5 effectively displays additional environment examples for the Emotion, Celebrity, and Living World categories as described in the caption. The visual layout is clear, with distinct color-coded labels for each category and legible text instructions for the tasks.

Figure 6

Figure 6 effectively displays additional environment examples for the Time, Traffic, and Public info categories. The visual layout is clear, with distinct labels for each category and specific instructions for the agent, fully supporting the caption's description.

Figure 7

Figure 7 effectively displays diverse environment examples for the specified categories (Attribute, State, Color, Symmetry, Shape, Counting) with clear visual labels and corresponding natural language instructions. The layout is organized and the content aligns perfectly with the caption's description of additional Act2Answer environment examples.

Action items.

- [**science**] Figure 1: The caption claims results for 'seven state-of-the-art VLA models', but the legend only lists five models (SpatialVLA, pi0, Magma, Xiaomi-Robotics-R0, InternVLA-M1).
- [**science**] Figure 1: The bottom panel displays results for 'Act2Answer' and 'Libero' tasks, but the radar chart (top panel) lacks a corresponding legend or label indicating which task suite the radial data represents.
- [**writing**] Figure 1: The bottom panel's 'Act2Answer' bars lack explicit percentage labels, unlike the 'Libero' bars, making precise comparison difficult.
- [**science**] Figure 4: The legend defines 'Parts' as PREFIX (solid line) and ACTION (dashed line), but the plot contains multiple lines of the same color and style (e.g., multiple solid blue lines) without distinguishing which specific model each line represents, making the data uninterpretable.
- [**writing**] Figure 4: The legend lists six models (SmolVLA, pi0, OpenVLA, SpatialVLA, Magma, Xiaomi-Robotics-R0) but does not explicitly map the specific line styles (solid vs. dashed) to the 'Prefix' vs. 'Action' components for each model, creating ambiguity in reading the graph.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on field-specific acronyms and jargon that are not consistently defined for a general audience, potentially excluding non-specialist readers.

First, the abstract introduces "VLA" (Vision-Language-Action) and "VLM" (Vision-Language Model) without defining them. While standard in robotics/AI, a general reader needs these spelled out immediately. The same applies to "SFT" and "RL" in Section 6.2; these acronyms appear without definition.

Second, the term "Action Expert" is used in Section 5.2 ("VLM and Action Expert parts") without explanation. It is unclear if this refers to a specific architectural component, a head, or a module. This internal terminology should be clarified (e.g., "the action-prediction head").

Third, the metric "Soft Success Rate" is abbreviated as "SR" in Section 5.1. While the full name is given once, the subsequent heavy use of "SR" in equations and text assumes the reader has memorized the abbreviation. It would be clearer to use the full term or a more descriptive variable name in the text.

Finally, terms like "backbone" (used frequently to refer to the base model) and "probe" (in "linear probe") are standard ML jargon. While acceptable in a specialized venue, the paper claims to address "commonsense" and "world knowledge," suggesting a broader relevance. Replacing "backbone" with "base model" and explicitly defining "linear probe" as "a simple linear classifier trained on..." would improve accessibility without sacrificing precision.

Action items.

- [**writing**] Define 'VLA' and 'VLM' at first use in the Abstract. Currently, 'VLA' appears in the first sentence and 'VLM' in the second without prior definition, which excludes non-specialist readers.
- [**writing**] Replace the acronym 'SR' with 'Soft Success Rate' or 'success rate' in Section 5.1. The text defines 'Soft Success Rate (SR)' but then immediately uses 'SR' in equations and subsequent text without re-stating the full term, creating a barrier for readers unfamiliar with the specific metric notation.
- [**writing**] Define 'Action Expert' in Section 5.2. The term is introduced as 'VLM and Action Expert parts' without explaining what an 'Action Expert' is (e.g., a specific head, a separate network, or a module). This is internal jargon that needs a plain-English explanation.
- [**writing**] Replace 'backbone' with 'underlying neural network' or 'base model' in Section 5.2 and throughout. While common in ML, 'backbone' is jargon that may confuse readers from other fields. Similarly, define 'probe' in the context of 'linear probe' (e.g., 'a simple linear classifier trained to...').
- [**writing**] Define 'SFT' and 'RL' in Section 6.2. The text mentions 'fine-tune the model with both SFT and RL' without defining these acronyms (Supervised Fine-Tuning and Reinforcement Learning). These are standard but should be defined for a general audience.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument for the need to evaluate knowledge retention in VLAs, and the proposed Act2Answer protocol follows logically from the identified gap in current benchmarks. The decomposition of task success into perception, knowledge, control, and environment (Section 4.1) provides a sound theoretical basis for the experimental design.

However, there are minor logical inconsistencies between the textual claims and the reported data in the results section. Specifically, in Section 5.2 (RQ2), the assertion that "no evaluated VLA reaches above-random performance on Symmetry or Counting" is technically contradicted by Table 1, where SpatialVLA and Magma achieve 52% on Counting. While 52% may not be statistically significant given the defined Delta (~5.7%), the phrasing "no evaluated VLA" suggests a strict upper bound of 50% or chance, which the data does not support. The text should be refined to state that no model *significantly* exceeds the chance threshold (0.5 + Delta) rather than implying no model scored above 50%.

Additionally, the claim in RQ2 that Magma is the "only clear exception" to near-chance performance on Emotion, Attribute, State, and Time is slightly overstated. While Magma performs best, other models like Xiaomi-Robotics-R0 (63% on Emotion) and InternVLA-M1 (58% on Living World) show marginal above-chance performance in specific categories. The text should qualify this claim to avoid implying a binary distinction where a spectrum of performance exists.

Finally, in RQ3, the comparison of VLMs to VLAs as "counterparts" is not always logically tight. For example, Paligemma-3B is listed alongside OpenVLA, but Paligemma is not the explicit backbone of OpenVLA. The conclusion that the performance gap is due to the VLM-to-VLA transition is valid in general, but the specific pairing in the table implies a direct lineage that is not always present, potentially confusing the causal attribution for those specific rows. Clarifying the backbone relationships or generalizing the comparison would strengthen the logical consistency of the argument.

Action items.

- [**writing**] In Section 5.2 (RQ2), the claim 'no evaluated VLA reaches above-random performance on Symmetry or Counting' contradicts Table 1 where SpatialVLA and Magma score 52% on Counting. Clarify that no model exceeds the statistical significance threshold (0.5 + Delta) rather than implying all scores are <= 50%.
- [**writing**] In Section 5.2 (RQ2), stating Magma is the 'only clear exception' to near-chance performance is an overgeneralization. Models like Xiaomi-Robotics-R0 (63% Emotion) and InternVLA-M1 (58% Living World) show marginal above-chance results. Qualify the claim to reflect a performance spectrum.
- [**science**] In Section 5.3 (RQ3), the text compares VLMs to VLAs as 'counterparts' (e.g., Paligemma vs OpenVLA) without establishing a direct backbone lineage for all pairs. This weakens the causal claim that the gap is solely due to the VLM-to-VLA transition for those specific rows. Clarify relationships or generalize the comparison.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the nature of knowledge retention in Vision-Language-Action (VLA) models that extend beyond what the provided data and methodology strictly justify. While the Act2Answer protocol is a valuable contribution, the interpretation of the results frequently conflates correlation with causation and attributes performance drops to "knowledge loss" without fully ruling out alternative explanations related to the evaluation protocol itself.

First, the abstract and introduction repeatedly claim that the protocol "reduces confounds from low-level control" and yields an "action-grounded success rate with reduced control confounds." This is an over-claim. The protocol requires the agent to perform a precise placement action (moving a cube to a specific tile). While this is simpler than long-horizon manipulation, it is not free of control requirements. The results in Table 1 show that models like π_0 and OpenVLA perform near chance on categories like "Symmetry" and "Counting." The paper attributes this to a lack of knowledge. However, it is equally plausible that these models fail due to the specific motor precision required to distinguish between two closely spaced tiles, or due to a failure in the visual grounding of the "left/right" spatial instruction, rather than a lack of abstract knowledge about symmetry. The text should be revised to explicitly state that while control complexity is *reduced*, it is not *eliminated*, and that low success rates cannot be definitively attributed solely to knowledge forgetting without further ablation studies isolating motor failure.

Second, the claim in the abstract and RQ5 that "VQA co-training is associated with better knowledge retention" presents a correlational finding as a causal mechanism. The study compares models like Magma (which has VQA co-training) against OpenVLA (which does not). However, these models differ significantly in architecture, pretraining data scale, and specific fine-tuning procedures. Attributing the performance gap primarily to the presence of VQA co-training ignores these confounding variables. The paper does not perform a controlled experiment (e.g., fine-tuning the same base VLM with and without VQA data) to isolate this variable. The language should be softened to reflect that VQA co-training is *correlated* with better performance in this specific sample, rather than being a proven driver of retention.

Finally, the interpretation of the layerwise intent probing results (RQ4) over-extrapolates from the limitations of linear probes. The paper concludes that "answer-relevant signals peak in middle VLA layers but attenuate in upper layers," suggesting a specific bottleneck where knowledge is lost or suppressed. However, linear probes only measure linearly separable information. A drop in probe accuracy in upper layers could simply indicate that the information is encoded in a non-linear, task-specific, or distributed manner that is optimal for action generation but invisible to a linear classifier. The paper does not rule out the possibility that the information is still present but transformed. The conclusion should be restricted to the recoverability of information by linear probes, rather than making definitive claims about the "attenuation" or "loss" of the signal within the model's internal representations.

Action items.

- **[writing]** The claim that Act2Answer 'reduces confounds from low-level control' is overstated. Failures in categories like 'Symmetry' may reflect motor precision limits rather than knowledge loss. The text should acknowledge that control confounds are reduced but not eliminated, and low SR could stem from motor errors.
- **[writing]** The claim that 'VQA co-training is associated with better knowledge retention' over-extrapolates from a small, non-experimental sample. The study compares existing models with different training histories without controlling for architecture or data scale. This is correlational, not causal. Language should be softened to reflect correlation only.
- **[writing]** The assertion that 'answer-relevant signals... attenuate in upper layers' implies a specific bottleneck. However, linear probing only shows information is less linearly recoverable, not that it is lost. The signal may be encoded non-linearly for action. The interpretation should be limited to probe recoverability, not signal attenuation.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper introduces Act2Answer, a benchmark for evaluating commonsense and world knowledge in Vision-Language-Action (VLA) models. From a safety and ethics perspective, the work is generally sound as it focuses on evaluation rather than deploying potentially harmful systems. However, several areas require clarification to ensure ethical rigor and mitigate potential biases.

First, the 'Celebrity' category (Section 4.2, Table 1) relies on recognizing public figures. The paper does not specify whether these figures are from a specific cultural context or if the dataset includes diverse global figures. Without this, the benchmark may inadvertently bias evaluation toward Western-centric knowledge or exclude non-public figures, raising fairness and representational bias concerns. The authors should clarify the cultural scope and diversity of the celebrity dataset.

Second, the 'Living World' category (Section 4.2, Table 1) involves distinguishing living from non-living entities. The paper does not mention if the dataset includes sensitive biological content (e.g., injured animals, human faces) or if ethical guidelines were followed for image selection. Clarify if human/animal subjects were used and if consent or ethical review was obtained, especially if real-world images were sourced from public datasets.

Third, the 'Emotion' category (Section 4.2, Table 1) uses images of human faces to test emotion recognition. The paper does not address potential biases in emotion datasets (e.g., gender, race, age) or the ethical implications of training/evaluating models on facial emotion data.

Add a statement on dataset bias mitigation or ethical considerations for emotion recognition, as this is a known area of concern in AI ethics.

Overall, the paper's focus on evaluation is positive, but addressing these ethical considerations will strengthen its rigor and ensure responsible AI development.

Action items.

- [**writing**] The 'Celebrity' category (Section 4.2, Table 1) relies on recognizing public figures. The paper does not state whether these figures are from a specific cultural context or if the dataset includes diverse global figures. Without this, the benchmark may inadvertently bias evaluation toward Western-centric knowledge or exclude non-public figures, raising fairness and representational bias concerns.
- [**writing**] The 'Living World' category (Section 4.2, Table 1) involves distinguishing living from non-living entities. The paper does not mention if the dataset includes sensitive biological content (e.g., injured animals, human faces) or if ethical guidelines were followed for image selection. Clarify if human/animal subjects were used and if consent or ethical review was obtained.
- [**writing**] The 'Emotion' category (Section 4.2, Table 1) uses images of human faces to test emotion recognition. The paper does not address potential biases in emotion datasets (e.g., gender, race, age) or the ethical implications of training/evaluating models on facial emotion data. Add a statement on dataset bias mitigation or ethical considerations for emotion recognition.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a well-structured empirical study with a clear hypothesis: VLA fine-tuning degrades commonsense knowledge retention. The sample size is substantial (1,720 unique items, 3,440 episodes), and the use of left/right swapping to control for positional bias is a strong methodological choice that enhances the robustness of the success rate metric. The decomposition of task success into perception, knowledge, and control is conceptually sound and helps isolate the specific failure mode (knowledge loss vs. control failure).

However, the statistical evidence supporting the central claims requires tighter presentation. While Appendix A defines the chance margin Δ (approx. 5.7% for N=300), the main text frequently describes results as "near chance" or "at random threshold" without explicitly stating whether the observed means fall outside the $0.5 \pm \Delta$ interval. For instance, the claim that "no evaluated VLA reaches above-random performance on Symmetry" is critical; if the best score is 52%, it is within the noise margin of the 50% baseline. The authors should explicitly report the statistical significance (or lack thereof) for these borderline cases in the main results table or text to avoid over-interpreting noise as a trend.

Furthermore, the linear probing analysis (RQ4) is compelling but lacks a crucial detail regarding data leakage. The text states probes are trained on "hidden states from all tokens of every transformer layer" for "each episode." If the probe is trained and evaluated on the same 1,720 items used for the main benchmark, the high "Retention" scores (e.g., 75% for Magma) may reflect overfitting to the specific dataset rather than a generalizable internal representation of the knowledge. The authors must clarify if the probe training set is disjoint from the evaluation set or if cross-validation was used. Without this, the claim that "information remains internally represented" is weakened by the possibility of memorization.

Finally, the mitigation experiments (Discussion) are based on a very small subset of categories and a single model. While the direction of the results is interesting, the sample size is too small to support the broad conclusion that "lightweight interventions are useful but limited." The evidence here is suggestive but not yet robust enough to generalize to the entire Act2Answer suite.

Action items.

- [**science**] The statistical interpretation of 'near-chance' performance requires explicit reporting of the calculated Delta (Δ) values for each category in the main text or a summary table, rather than only in the appendix. Currently, readers cannot verify if specific model scores (e.g., 52% on Symmetry) are statistically distinguishable from 50% without manual calculation.
- [**science**] The claim that 'no evaluated VLA reaches above-random performance on Symmetry or Counting' relies on aggregated scores. The paper should explicitly report the standard error or confidence intervals for these specific category averages to confirm that the mean is not significantly different from 0.5, given the N=300 sample size.
- [**science**] The linear intent probing results (Figure 4, Table 3) show high retention in intermediate layers but low action success. The paper should clarify if the probe training set overlaps with the evaluation set. If the probe is trained on the same 1,720 items used for evaluation, the 'retention' metric may be overfitting to the specific dataset rather than measuring generalizable internal representations.
- [**science**] The mitigation experiments (Table 10) use a very small subset of categories (Emotion, Attribute, Color, Shape) and a single model (0). The conclusion that 'lightweight interventions are useful but limited' is based on insufficient statistical power. The authors should either expand the ablation to more categories/models or temper the claim to reflect the limited scope of the evidence.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in this paper is generally sound in its conceptual approach, particularly the definition of the Soft Success Rate (SR) and the use of a chance margin (Δ) to distinguish signal from noise in a binary-choice setting. The decision to swap left/right configurations to control for positional bias is a strong methodological choice that improves the validity of the success rate metric.

However, there is a critical statistical flaw in the calculation of the significance threshold Δ described in Appendix \ref{appendix:delta}. The authors define Δ using the Wald confidence interval formula for a binomial proportion, assuming $N = 300$ independent trials for most categories. This assumes that the 300 episodes (150 unique items evaluated in two spatial configurations) are independent. This assumption is violated because the two evaluations of the same item are highly correlated; they share the same visual content, instruction, and underlying model state, differing only in the spatial permutation. Treating these as independent trials artificially inflates the effective sample size, leading to an underestimation of the standard error and a Δ that is too narrow. Consequently, the authors may incorrectly classify results as "statistically distinguishable from chance" when they are not. The analysis should be repeated using $N = 150$ (the number of unique items) as the effective sample size, or a cluster-robust standard error approach should be employed.

Additionally, the "Chance-Normalized Retention" metric (Section 5.3, Eq. 10) relies on the maximum probe accuracy across layers. This "max-pooling" approach is sensitive to noise and does not provide a measure of uncertainty. Given that linear probing results can be noisy, reporting confidence intervals for the Retention metric or the underlying probe accuracies would strengthen the claims regarding where knowledge is preserved or lost.

Finally, while the paper frequently states that models perform "near chance" or "above chance" in the Results section (Section 5.2), it would be more rigorous to explicitly map these qualitative claims to the quantitative Δ thresholds calculated in the appendix. For instance, explicitly listing which categories for which models satisfy $SR > 0.5 + \Delta$ would remove ambiguity.

Action items.

- **[science]** Appendix 'Chance Margin Delta' treats 300 episodes (150 items x 2 swaps) as independent Bernoulli trials. Since swapped pairs are correlated, this violates independence, underestimating standard error. Recalculate Delta using N=150 (unique items) or use cluster-robust variance.
- **[science]** The 'Chance-Normalized Retention' metric (Eq. 10) uses max-pooling over layers, ignoring variance and probe uncertainty. Please report confidence intervals for this metric or justify why point estimates suffice given probing noise.
- **[writing]** In Section 5.2, claims of 'near chance' performance lack explicit statistical backing. Please explicitly list which (model, category) pairs satisfy |SR - 0.5| <= Delta based on the appendix calculations rather than relying on qualitative descriptions.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a clear and engaging narrative regarding the evaluation of commonsense knowledge in Vision-Language-Action (VLA) models. The writing is generally fluent, and the logical flow from the problem statement to the proposed Act2Answer protocol is well-structured. The abstract effectively summarizes the contributions, and the introduction successfully motivates the research gap.

However, there are several instances of grammatical errors and stylistic inconsistencies that slightly impede readability. In Section 3.2 ("Commonsense Knowledge"), the opening sentence ("What matters for embodied agents is not knowledge in the abstract, the question is what kinds...") is a comma splice that requires punctuation correction. Similarly, in Section 5 ("Evaluation and Results"), a comma splice appears in the description of the experimental setup regarding OpenVLA exceptions.

Additionally, the capitalization of knowledge categories is inconsistent. In the Introduction (Section 1, Contribution 2), the list of categories (attribute, state, color, etc.) is presented in lowercase, whereas in Section 3.2, these same categories are introduced as bolded, capitalized headers (Physical, Temporal, etc.). For consistency and clarity, the list in the Introduction should match the capitalization style used in the detailed definitions or the tables.

Finally, some sentences are overly long and complex, particularly in the methodology descriptions. For instance, in Section 4.1, the description of the protocol includes a parenthetical figure reference that interrupts the sentence flow. Integrating the figure reference more smoothly would improve the reading experience. Addressing these minor grammatical and stylistic issues will enhance the overall polish of the paper.

Action items.

- [**writing**] In Section 3.2, the sentence 'What matters for embodied agents is not knowledge in the abstract, the question is what kinds...' contains a comma splice. Split into two sentences or use a semicolon.
- [**writing**] In Section 1, the list of categories (attribute, state, color...) uses lowercase, while Section 3.2 capitalizes them. Standardize capitalization for consistency across the manuscript.
- [**writing**] In Section 4.1, the parenthetical figure reference breaks the sentence flow. Integrate the reference smoothly, e.g., '...as shown in Figure X'.
- [**writing**] In Section 5, the sentence regarding OpenVLA exceptions contains a comma splice. Use a semicolon or rephrase to 'with the only exception being...'.